

Extraocular Muscle Repositioning and Diplopia

Associated with Ophthalmic Plaque Radiation Therapy for Choroidal Melanoma

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Objective: To evaluate extraocular muscle surgery associated with plaque brachytherapy for choroidal melanoma.

Design: Single-center retrospective cohort study.

Participants: Three hundred twenty-nine eyes of 329 consecutive patients with clinically diagnosed choroidal melanoma.

Intervention: Palladium 103 plaque brachytherapy with or without extraocular muscle surgery.

Main Outcome Measures: Type of muscle surgery required for each tumor location, timing, incidence, and duration of diplopia, as well as treatment.

Results: Two hundred fifty-four patients (n = 254/329; 77.2%) required muscle surgery. One hundred seven patients (n = 107/329; 32.5%) required surgery on 2 or more muscles. Of 373 muscles repositioned, the lateral rectus muscle (n = 115/373; 30.8%) and inferior oblique muscle (n = 70/373; 18.7%) were the most common, correlating to intraocular tumor location ($P < 0.001$). Only 6 tumors (n = 6/61; 9.5%) originating from the iris and ciliary body required muscle surgery for plaque placement. Of the 312 patients with a preoperative visual acuity better than 20/400, diplopia was reported at the first postoperative visit by 41 patients (n = 41/312; 13.1%), 2 of whom had not undergone muscle surgery. Diplopia resolved spontaneously within 1 month in 18 patients (n = 18/41; 43.9%), between 1 and 6 months in 12 patients (n = 12/41; 29.3%), and at more than 6 months in 5 patients (n = 5/41; 12.2%). Among the 312 patients, persistent diplopia occurred in 6 patients (1.9%), including 1 who had not undergone muscle surgery. Treatment was declined in 1 patient, 3 patients (n = 3/41; 7.3%) were treated with prisms, and 2 patients (n = 2/41; 4.9%) required surgery.

Conclusions: Extraocular muscle surgery frequently is required for plaque brachytherapy. Although transient diplopia occurred in 11.2% of patients, persistent diplopia occurred in only 1.9% of patients and was treatable. *Ophthalmology* 2014;121:2268-2274 © 2014 by the American Academy of Ophthalmology.

Plaque brachytherapy is the most widely used eye- and vision-sparing treatment for choroidal melanoma.¹⁻⁷ This is due (in part) to the Collaborative Ocular Melanoma Study finding that iodine 125 (I^{125}) plaque irradiation was equivalent to enucleation for survival of patients with select medium choroidal melanomas.⁶ Also supported by 2014 guidelines issued by the American Brachytherapy Society (ABS), universal consensus has suggested that plaque therapy can be used for almost all iris, ciliary body, and choroidal melanomas.⁷

The ability of treatment to destroy the primary tumor is critical to successful cancer therapy. For uveal melanoma, failure of local control typically results in secondary treatments (laser, irradiation, or enucleation) as well as a greater risk of vision loss and metastatic disease.^{8,9} Failure of local control has been related to incorrect plaque placement.^{1,4,5,10,11} Correct plaque placement is defined as covering the tumor's base, plus a 2- to 3-mm surround of

normal-appearing tissue.⁷ This often requires repositioning and temporary displacement of selected extraocular muscles.

Diplopia and cyclotorsional symptoms have been reported after plaque radiation therapy.¹²⁻¹⁵ Similar to diplopia seen after cataract surgery (incidence, 0.21%–0.98%), retinal detachment surgery (incidence, 3%–14%), and glaucoma surgery (incidence, 1.4%–5%), postoperative strabismus after plaque surgery is likely the result of direct muscle damage, displacement, inflammation, and scarring.¹⁶⁻²¹ Although treatment typically is not needed for orthophoric patients, diplopia may persist and require corrective prisms or surgery.

Our study examined the complications of muscle repositioning in ophthalmic plaque radiation therapy for uveal melanoma. We present data on commonly affected muscles, the incidence of symptomatic postoperative diplopia, and its corrective treatment.